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Creativity as a distinct trainable mental state: An EEG study of musical improvisation

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ABSTRACT

Alpha-band EEG was used to index how creative mental states relate to the creation of artistic works in skilled musicians. We contrasted differences in frontal upper alpha-band activity between tasks with high and low creativity demands by recording EEGs while skilled musicians listened to, played back, and improvised jazz melodies. Neural responses were compared for skilled musicians with training in musical improvisation versus those who had no formal improvisation training. Consistent with our hypotheses, individuals showed increased frontal upper alpha-band activity during more creative tasks (i.e., improvisation) compared to during less creative tasks (i.e., rote playback). Moreover, this effect was greatest for musicians with formal improvisation training. The strength of this effect also appeared to modulate the quality of these improvisations, as evidenced by significant correlations between upper alpha EEG power and objective post-hoc ratings of individuals' performances. These findings support a conceptualization of creativity as a distinct mental state and suggest spontaneous processing capacity is better nurtured through formal institutional training than informal.

In an address to the American Psychological Association, Guilford (1950) suggested that creativity is an aptitude that is psychometrically measurable and worthy of study. An ongoing challenge in cognitive science has been the ability to accurately quantify this ability. Over the past half-century researchers have conceptualized creativity in several ways, defining the construct as a cognitive ability, as a value assigned to products, and as a mental state (i.e., mode of being; Albert and Runco, 1999). In 1974, Torrance developed the Torrance Tests of Creative Thinking (TTCT; Torrance, 1974), aimed at measuring various creative abilities, thereby assigning normative values to an individual's capacity to think in novel and original ways. This gave rise to a cognitive ability conceptualization of creativity, where individuals can be labeled as being of low, average, or high creativity. Although arguably successful in measuring creative aptitudes, cognitive ability tests such as the TTCT do not include created products as indicators of creative abilities. Several researchers (Amabile, 1983; Mackinnon, 1978; Runco, 1989a; Taylor, 1960; Treffinger and Poggio, 1972; Wallach, 1976) have responded to this omission by broadening the assessment of creativity to include artifacts (i.e., evidence of creativity). This second wave of researchers located creativity as embedded in the creative product and they advocated for the adjudication of samples of actual creative work by domain experts as essential to the determination of creativity. Here, a person can be labeled as more or less creative as a function of the degree of uniqueness, originality, and appropriateness inherent in their work.

Shifting perspective, artists themselves often conceptualize creativity experientially (Berkowitz, 2010; Berliner, 1994; Nardone, 1997; Nelson and Rawlings, 2007) – as a mode of being at a given moment in time. Aligning with this experiential conceptualization of creativity is Csikszentmihalyi's (1996) well-known concept of *flow*. Csikszentmihalyi conceptualized creativity and flow as two separate phenomena, defining creativity as the creation of novelty, and flow as the mental state wherein individuals experience "an almost automatic, effortless, yet highly focused state of consciousness" that workers experience while performing a task (Csikszentmihalyi, 1996, p. 110). Per this theory, any task can be an opportunity for flow if the parameters for flow are followed or are present.

Along similar lines, Green (2016) conceptualized creativity as a mental state, characterized by the intent to generate novel ideas, and used neuroscientific methods (e.g., fMRI) to test this conceptualization (Green et al., 2015). On this view, individuals augment their state creativity by deliberately trying to think more creatively. Green relates augmenting one's creative state to the colloquially used concept of putting on one's "thinking cap". To this end, he presents several findings supporting a state-based conceptualization (Green et al., 2012; Green et al., 2015), however, his concept of a creative mental state differs from ours.

In the present study, we conceptualize creativity as a distinct mental state that includes (a) engagement in an activity, (b) spontaneous

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processing of thoughts, and (c) the expression of these thoughts through a medium (e.g., voice, a writing tool, a musical instrument). For example, during jazz pianist Keith Jarrett's (1975) Koln concert, Jarrett was engaged in playing piano, spontaneously composing music by channeling a stream of intuitive thoughts, and expressing these thoughts through his piano in real time. In our proposed state-based conceptualization, *being* is used as a verb and *creative* as a modifier describing the way a person is being during an interval in time. In this sense, when a person is engaged in a creative activity that person is being creative, if these parameters are involved. Based on the above explanation, it can be stated that for the duration of the Koln concert, Keith Jarrett was being creative. Thus, the aim of the present work is to examine this operationalization of creativity in an objective way.

An advantage of both our and Csikszentmihalyi's conceptualizations is that they align more closely with artists' reported experiences of creativity than do the ability and product orientations, however the intangible nature of experiential conceptualizations such as these makes them difficult to study objectively. Can creativity be conceptualized as a distinct mental state or is such a conceptualization too immaterial?

In the current study, we aimed to experimentally isolate a neural marker of this creative mental state by comparing neural activity during spontaneous versus rehearsed artistic activity. Specifically, we used electroencephalography (EEG) to monitor the frontal brain areas of musicians while being creative compared to while not. Further, a design was used in which we observed neural markers of this creative state in individuals trained in musical improvisation and musicians of a similar skill level who had no formal improvisation training. In so doing, we assumed that differences in task-related neural responses reflected underlying changes in cognitive processing modes. The spontaneous composition and rehearsed playback tasks we used were intended to resemble naturalistic tasks of high and low creativity. This assumption is supported with theories on the neurocognitive mechanisms supporting mental states (i.e., cognitive processing modes) associated with creativity.

1. Creative processing models

One way to conceptualize thought processes during creativity is through the interplay of executive control, long-term memory and working memory. Working memory has been conceptualized as the site of conscious awareness (Courtney et al., 1998; Dietrich, 2004a), however the way information manifests into working memory differs as a function of cognitive processing mode. Dietrich (2004a) advanced a spontaneous processing model delineating two competing modes occurring during cognitive ideation—a deliberate mode and a spontaneous mode. The deliberate mode maps to what is sometimes referred to as *analytical* processing, whereas the spontaneous mode relates instead to *intuitive* processing (Shirley and Langan-Fox, 1996; Simonton, 1975). In the deliberate processing mode, attention guides the retrieval of information stored in long-term memory whereas in the spontaneous processing mode attention is less deliberate, allowing for intuitive thoughts to surface into conscious awareness. In deliberate processing, the dorsolateral prefrontal cortex, hypothesized to be recruited in actively retrieving specific information and maintaining it in short-term memory for a transient duration (Cabeza and Nyberg, 2000; Hasegawa et al., 1999), instigates controlled information retrieval from long-term memory (Friedman and Förster, 2002). Conversely, in spontaneous processing the attention system does not deliberately select the content of working memory; instead unguided thoughts enter awareness as a stream of ideas. From this perspective, disengaging the dorsolateral prefrontal cortex from deliberate processing allows unpremeditated ideas to surface (Dietrich, 2003). This spontaneous processing mode can therefore be conceptualized as characterizing a distinct mental state marked by the manifestation of intuitive thoughts in working memory and occurring when the frontal attentional network is disengaged from deliberately controlling the content of the working memory buffer. As such, the nature of information in working memory at any given moment differs depending on whether semantic information is retrieved by deliberate processing or whether intuitive thoughts are allowed to emerge into consciousness via spontaneous processing. Deliberate processing yields thoughts that are structured and conform to an internalized sense of what will be deemed socially acceptable or worthy, whereas spontaneous processing tends to yield an intuitive stream of unfiltered, non-conforming thoughts.

In another line of inquiry, researchers have framed these deliberate and spontaneous modes in terms of a dual-process model of creative thinking. Whereas Stanovich (1999) delineated between intuitive *System 1* and reflective *System 2* processing, Evans, (2008, 2009) refined and renamed these modes, referring to them as *Type 1* and *Type 2* processes. Type 1 processes correspond to *gut reactions*, whereas Type 2 correspond to more reflective, analytical, and evaluative thinking. Type 1 processes are fast and automatic, bypassing conscious deliberation and effortful thinking whereas Type 2 processes are slower, involving conscious deliberation and effortful recall of ideas from memory (Sowden et al., 2015). Thus, Type 1 processes are analogous to what Dietrich (2004a) described as a spontaneous mode and Type 2 to a deliberate mode. Although both Type 1 and Type 2 processing are likely important in creative thinking, the more intuitive Type 1 processing may be especially valuable during improvisation in the performance arts when ideation and response speed for artistic expression are prioritized. During improvisation, Type 2 deliberation and analytical thinking may simply be too slow and too constrained by rules and inhibitions (e.g., perceptions of social evaluations).

To summarize these models, creative thinking varies as a function of processing mode, with one mode calling for deliberate, controlled retrieval in generating ideas, and the other for less deliberate, less controlled streaming of ideas into the working memory buffer. These processing modes align with individuals' experiences of creativity (Berkowitz, 2007; Berliner, 1994; Nardone, 1997; Nelson and Rawlings, 2007) and suggest distinct mental states for more and less spontaneous creative tasks. Collectively, the above-mentioned work has posited a competitive processing mode interaction theory, however, in light of recent findings from functional magnetic resonance imaging (fMRI) and transcranial direct current stimulation (tDCS), an alternative concurrent interaction theory has emerged. Below, we review this work.

2. fMRI and tDCS research on generative artistic performance

Several researchers have investigated the neural bases of musical improvisation using fMRI, with some interpreting activation patterns in specific brain areas as signifying that musicians approach improvisation through either spontaneous or deliberate (i.e., Type 1 vs. Type 2) creative processes. For example, in a conjunction analysis showing activity common to both melodic and rhythmic improvisation, Berkowitz and Ansari (2008) found activation in the left inferior frontal gyrus (IFG), dorsal premotor cortex (PMD), and rostral anterior cingulate cortex (ACC). The left IFG and the ACC have been implicated in deliberate recall processes such as controlled memory retrieval (Badre et al., 2005; Hirshorn and Thompson-Schill, 2006), voluntary selection, and evaluative decision making (Forstmann et al., 2006; Walton et al., 2004) –processes reminiscent of Type 2 creative thinking. Conversely, Limb and Braun (2008) asked musicians to play both pre-learned melodies and free improvisations on an fMRI-friendly keyboard and found deactivation of the dorsolateral prefrontal cortex (DLPFC) and activation of the medial prefrontal cortex (MPFC) during improvisation. In research on mind-wandering states, the brain's default mode network regions were associated with attenuated control of memory retrieval, and with attenuated selection of the content of working memory (Andrews-Hanna, 2012). Along these lines, Limb and Braun interpreted the

pattern of activity they observed during improvisation as a transient suspension of the executive functions deliberately controlling memory retrieval in favour of less deliberate creative processes occurring during improvisation. In sum, these seminal studies support a theory of competing frontal activation patterns for different processing modes, with frontal activation related to Type 2 deliberate and frontal deactivation related to Type 1 spontaneous processing.

More recently, Pinho et al. (2016) investigated the interplay of large-scale neural networks involved in artistic performance. During a musical performance task biased towards generative Type 1 ideation, Pinho et al. found increased connectivity between the DLPFC and the default network–a network associated with diffuse, spontaneous, and internally focused attention (Andrews-Hanna, 2012). Conversely, during a musical performance task biased towards deliberate Type 2 ideation, they observed increased functional connectivity of the DLPFC and the control network–a network associated with more externally focused attention (Seeley et al., 2007). In summary, Pinho et al. contrasted tasks that elicited a competitive interplay of deliberate and spontaneous processing modes echoing Berkowitz and Ansari (2008), and Limb and Braun's (2008) observations.

It is plausible, however, that a view of Type 1 processes as unilaterally underlying spontaneous artistic performance is overly simplistic. Creativity itself is often characterized by appropriateness or goodness as well as novelty, and it follows to assume that when artists improvise they to some degree evaluate and select, or at least monitor, the stream of thoughts they generate. In a pair of studies of the networks involved in visual arts (Ellamil et al., 2012) and in poetry (Liu et al., 2015), an inverse correlation between default and control network activity during Type 1 creative thinking was observed. However, in more goal-directed tasks biased towards Type 2 creative thinking, both default and control networks were implicated in a *functional coupling* of these networks. Beaty et al. (2016) synthesized these fMRI studies and suggested an interplay of generative and evaluative processes that can occur either competitively or concurrently as modulated by the degree to which the creative tasks are goal-directed. Specifically, Beaty et al. explained that goal-directed artistic performance involves evaluative monitoring and selection, signifying activation of the control network. Conversely, less goal-directed artistic performance involves less evaluative monitoring and thus benefits from control network inhibition and increased reliance on the more diffuse attention provided by the default network.

The interplay of creative modes and the networks involved may also be modulated by expertise. In research testing the effects of transcranial direct current stimulation (tDCS) on improvised performance quality in expert and novice jazz musicians, Rosen et al. (2016) observed that anodal stimulation (i.e., application of electrical current to increase cortical excitability) applied to a scalp site over the right dorsolateral prefrontal cortex (*r*-DLPFC) resulted in performance gains amongst novice but not expert improvisers. Rosen et al. interpreted these results as "evidence supporting a dual-process creativity model in which the recruitment of Type-1 and Type-2 processes differs for experts and non-experts" (p.10). Specifically, Rosen et al. interpreted that the performance gains they observed were related to better Type 2 deliberate processing for novices whereas such processing was detrimental for experts who rely predominately on Type 1 generative processing. These results suggest that novice improvisers rely on top-down executive control in generating high quality improvisations, whereas experts do not.

In sum, fMRI and tDCS research over the last decade suggests generative artistic performance is related to an interplay between deliberate and spontaneous processing modes, and that the networks involved in these modes may be modulated by degree of task goal-direction and by expertise. In the current study, we build on this emerging framework of the neural bases underlying spontaneous and deliberate processing by contrasting naturalistic musical performance tasks biased towards Type 1 and Type 2 creative thinking while monitoring frontal brain areas using EEG. We chose EEG as a complement to the available fMRI and tDCS evidence described above as well as for its high temporal accuracy in measuring state-related brain-wave changes. As such, in the next sections, we review EEG research on creative cognition.

3. EEG research on creativity

The underlying assumption of EEG in cognitive neuroscience is that post-synaptic potentials in the cerebral cortex are reflected in electrical potentials that can be detected at various sites across the scalp. These can be described as amplitude or power values in different spectral frequency ranges (e.g., alpha, beta, delta; Rippon, 2006). The alpha (8–13 Hz) and beta (14–30 Hz) ranges are most commonly associated with conscious wakeful activity, with alpha activity associated with relaxed states (e.g., altered states of consciousness, intuitive states), and beta activity associated with logical reasoning (e.g., rational thought, critical analysis; Carlson, 2007).

Recently, some work has emerged contrasting views on the sensitivity of alpha to creativity-related manipulations. In separate reviews, Dietrich and Kanso (2010), and Arden et al. (2010) highlighted substantial heterogeneity across results and concluded pessimistically. Conversely, Fink and Benedek (2014) attributed heterogeneity across results to inconsistent conceptualizations of creativity and to variation in the tasks and methods used across studies. Moreover, they made a compelling argument for the role of alpha in creative cognition stating that "the EEG alpha frequency band is sensitive to various creativity-related demands involved in creative ideation; thereby revealing a quite consistent and replicable picture about some promising brain mechanisms relevant for creativity" (p. 112).

In the sections below we review research on alpha during creative ideation in both domain-general and artistic domain research. Because of the prominent role attributed to the DLPFC in models of spontaneous processing, we focus on results involving frontal brain regions. Major strands include inquiry into task-related alpha, the relationship of alpha and originality, and the sensitivity of alpha to creativity inducing interventions. Whereas a substantial amount of research on alpha band sensitivity to domain-general creative cognition has accumulated, much less artistic domain research exists.

4. Alpha activity during creative cognition

EEG research has shown increased frontal alpha activity during divergent thinking (i.e., tasks involving open-ended questions and stimuli that have many possible answers) compared to during convergent thinking (i.e., tasks involving questions and stimuli that have only a single correct response) (Fink et al., 2007; Jauk et al., 2012). For example, Fink et al. (2007) found strong frontal alpha synchronization during creative ideation tasks (e.g., naming original uses for common household objects), whereas a more intelligence-related task (e.g., completing suffixes) elicited substantially less alpha synchronization. Fink and his colleagues interpreted their findings to show that the degree of creativity inherent in a task correlates with alpha synchronization in frontal brain regions. In accord with this interpretation, Jauk et al. (2012) found that divergent thinking tasks were accompanied by frontal alpha synchronization whereas convergent thinking tasks were accompanied by frontal alpha desynchronization. Taken together, the above findings indicate that task-related frontal alpha activity may be modulated by type of creative thinking (i.e., convergent or divergent thinking) with divergent thinking eliciting increased alpha band activity (Fink and Benedek, 2014).

A pair of studies from the same group (Fink et al., 2006, 2011) have also found that training interventions affect alpha synchronization during creative ideation. Fink et al. (2006) observed greater originality of ideas and stronger frontal alpha activity in a group of participants trained for two weeks on divergent thinking compared to a group that

did not receive training. Similarly, Fink et al. (2011) investigated alpha activity in response to very brief creativity-inducing interventions such as being exposed to other people's creative ideas and being presented with the sounds of other people's laughter. They found that those that received the interventions showed increased right frontal alpha activity during creative ideation tasks whereas those that did not receive the intervention did not show increased frontal alpha.

5. Alpha activity during creative cognition - artistic domains

A related question concerns whether EEG alpha effects also extend to creativity in artistic domains (for a review, see Dietrich and Kanso, 2010). Here the results are somewhat more equivocal but appear to lean towards increased frontal alpha during artistic improvisation. In studies involving the mental imagery of artistic tasks (e.g., spontaneous mental composition of non-rehearsed drawings on a white wall), Bhattacharya and Petsche, (2002, 2005) and Petsche et al. (1997) found frontal alpha desynchronization. In contrast, Fink et al. (2009b) found upper alpha (10–12 Hz) synchronization during spontaneous free-form dance. Similarly, Dikaya et al. (2016) more recently found frontal lower alpha (8–10.5 Hz) interhemispheric connections during visual art improvisation which they interpreted as reflecting top-down inhibition of task-irrelevant processes. In a recent study of musical improvisation, Adhikari et al. (2016) found alpha (8–12 Hz) synchronization during improvised piano performance. Adhikari et al.'s conclusions support the notion that creativity, and specifically musical improvisation, may be facilitated by generative bottom-up processing and attenuated top-down control–processes associated with frontal alpha activity (Fink and Benedek, 2014).

Finally, although there is a contemporary consensus that creativity is not associated with one hemisphere or the other, alpha synchronization effects during artistic creativity do show some evidence of lateralization. Multiple studies have shown increased alpha power in the right hemisphere (Fink et al., 2009b; Martindale et al., 1984), and increased right hemispheric synchrony in other frequency bands (Bhattacharya and Petsche, 2002, 2005; Petsche et al., 1997).

In sum, there are very few studies wherein researchers have monitored EEG alpha during artistic improvisation and, due to equivocal findings, it remains difficult to accurately specify the mechanisms by which certain creativity processes would involve synchronization vs. desynchronization. Moreover, the literature suggests that frontal brain areas play a role in both domain-general and artistic-domain creative cognition, however heterogeneity in these findings persist. Further, the relationship of alpha and originality in generated products, creativity-inducing interventions, and individual differences in personality variables remain largely unexplored. Much of the research has also examined the full alpha range (8–12 Hz), revealing opportunities for exploration of lower (8–10 Hz) and upper (10–12 Hz) bands respectively.

In the present work, we monitored upper alpha while contrasting tasks biased towards spontaneous and non-spontaneous processing, with a special focus on EEG changes at prefrontal sites. We further explored the role of ad-hoc creativity-inducing interventions (that is, formal improvisation training), relationships of alpha with the quality of improvised performances, and we isolated the upper alpha band as it has been shown to be sensitive to task manipulations (Fink and Benedek, 2014). Our aims and hypotheses are described below.

6. Aims and hypotheses

The first aim of this study was to investigate whether frontal oscillatory activity changes occurring during spontaneous musical creativity are different than during non-spontaneous (i.e., deliberate) musical creativity. We examined alpha synchronization in musicians during rote playback, free improvisation, and passive musical listening. While the rote playback task is analogous to a convergent thinking task, the free improvisation task is analogous to a divergent thinking task. Therefore, we expected to observe patterns of neural activity during improvisation like those found in previous studies on alpha activity during divergent thinking (Fink et al., 2007; Jauk et al., 2012; Jausovec, 1997; Krug et al., 2003) and during spontaneous artistic composition (Fink et al., 2009b; Martindale et al., 1984). Specifically, we hypothesized that EEGs would show increased alpha synchronization in frontal brain regions during spontaneous musical composition (i.e., improvisation) compared to during deliberate rote musical performance, and compared to when just listening to music. This finding would support the delineation of intuitive Type I and analytic Type II processing modes during creative ideation, specifically the notion that when accessing an intuitive stream of ideas people that are improvising are in what Dietrich (2004a) called a spontaneous mode. Further, differences in frontal brain activity during time intervals of spontaneous creativity would support our above-outlined experiential conceptualization of creativity, specifically that creativity is a distinct mental state characterized by specific cognitive processes.

Given prior findings showing that alpha activity increases as a result of creativity training (Fink et al., 2006), our second aim was to investigate the role a specific type of creativity training plays in neural manifestations of a creative state. This is addressed by observing musicians with formal institutional training in creativity and musicians without formal institutional training in creativity. Specifically, we recruited jazz and classical musicians both with and without formal institutional training involving improvisation. We hypothesized that musicians with formal institutional training in improvisation (FITI) would show increased frontal alpha activity compared to musicians without formal institutional training in improvisation (non-FITI). This finding would support Gagné's (2005) theory that skills are developed through formal institutional training.

The third aim of this study was to investigate whether alpha activity during spontaneous musical composition is associated with higher quality improvised musical performances as rated by domain experts. We further aimed to investigate whether this relationship would occur in groups with and without FITI.

A key design feature of the present experiment is the assessment of creativity to identify groupwise and individual differences in this regard. Prior work has attempted to measure the construct of creativity using a variety of measures including assessing creative processes, personality and behavioural correlates of creativity, attitudes towards creativity, environmental factors that foster creativity, and characteristics of creative products (Plucker and Makel, 2010). Here we adopt Runco (1989a), and Baer et al.'s (2004) proposal that product assessments represent an ideal measure of creativity. Such assessments typically include the rating of artifacts of the creative process by the creator or by external adjudicators. Although several instruments exist to assess creative products, including the Creative Product Semantic Scale (Besemer, 1998) and the Student Product Assessment Form (Reis and Renzulli, 1991), here we have selected the most commonly used of these, Amabile's (1982) Consensual Assessment Technique (Plucker and Makel, 2010). This technique is a loosely bound way of measuring the creativity of products based on judgments made by independent raters, and has been used extensively as a measure of between-subjects differences in creativity (Baer et al., 2004; Baer and Oldham, 2006; Hennessey, 1994; Kaufman et al., 2008). It is also known to be a reliable measure with a high degree of ecological validity (Hennessey et al., 2010).

7. Method

7.1. Participants

Musicians were recruited from The University of Western Ontario community and surrounding London Ontario region, via recruitment posters and word of mouth. A total of 22 people participated in this study (9 female). The mean age was 26.2 years (SD =9.5; range 18–54 years). The mean number of years of music experience (i.e., amount of experience playing music) was 18.5 years (SD =11.7) with the least experienced musicians having four years and the most experienced having 48 years. The mean number of years of improvisation experience (i.e., amount of experience improvising musically) was 10.2 years (SD =13.1) with the least experienced having no experience and the most experienced having 48 years. Participants were required to be 18 years of age or older, right-handed, and neurologically healthy by self-report. An additional nine participants were recruited but subsequently excluded from analyses due to excessive physiological noise during EEG (N =7), ambidexterity (N =1), or being a statistical outlier in one of the tasks (frontal alpha synchronization > 2SD above the mean in the right hemisphere during the improvisation task).

Building on Gagné's (2005) differentiated model of gifts and talents, the sample in the current study was split into two groups: those with previous formal institutional training in improvisation (*FITI*), and those without previous formal institutional training in improvisation (*non-FITI*). Previous formal institutional training, assessed using a demographic questionnaire, included organized programs and courses (e.g., university, college, high school, Royal Conservatory of Music, other privately organized programs and courses lasting four weeks and more), and consistent private lessons over an extended duration (four weeks and more). Previous non-formal training included being self-taught, and descriptions such as attending an occasional workshop, through books and articles, by hearing and watching others, and by experimenting.

There were 12 participants in the non-FITI group (six females; mean age 24.9 years, SD =8.2, range 18- 46 years). The mean number of years of overall music experience was 17.3 years (SD =11.4), ranging from 4 to 45 years. The mean number of years of improvisation experience was eight years (SD =12.4) with the least experienced musicians having no experience and the most experienced having 45 years. The FITI group consisted of 10 musicians, (three female; mean age 27.7 years. SD =11.1; range 19- 54 years). The mean music experience was 19.9 years (SD =12.5), ranging from 6 to 48 years. The mean improvisation experience was 12.8 years (SD =14.1), ranging from 2 to 48 years.

8. Experimental procedure

Participants provided demographic data as well as data on their musical and improvisational experience. EEG recording took place in a sound attenuated room. A two-octave MIDI keyboard was positioned in a manner that made it easily accessible for right-hand playing. Before commencing, participants were briefly shown music charts for the three 16-bar music progressions to be presented, and were told the diatonic structures native to each progression (e.g., C-blues scale, G-major scale, D-modal) These charts had notation of the chord progressions, but were devoid of melodic transcriptions.

Task conditions progressed in the following order for each of the three pieces: *Listen*: listening to the melody without playing; *Learn*: actively learning to play the melody on the keyboard; *Imagine playback*. imagining playing the melody "straight", i.e., as closely to what they heard during the listen condition as possible; *Actual playback*: physically playing back the melody as closely as possible to what they heard during the listen condition; *Imagine improvisation*: imagining improvising freely over the chord changes. *Actual improvisation*: physically improvising freely over the chord changes.

Each of these conditions was 32-bars (approx. 80 s) in duration, and participants progressed through the six conditions for each piece of music before proceeding to the next piece. For all the above conditions, participants were directed to play with right hand only. After explaining this procedure, participants were reminded to play with as little embellishment as possible during the playback conditions, and to be as creative and free as possible during the improvisation conditions. Fifteen seconds of pre-stimulus reference EEG activity was also recorded. During the pre-stimulus interval participants focused on a fixation cross that was presented on the computer screen in front of them. Visual and auditory stimuli were presented using E-prime 2.0 software (Psychology Software Tools, Pittsburgh, PA) using a 19-in. CRT monitor and a near-field studio monitor style speaker (five inch woofer). The entire procedure took approximately 18 min. Participants were compensated with $20 for participating.

9. EEG procedures

9.1. EEG acquisition

Data were amplified at 500 Hz, at a gain of 500 using a SynAmps amplifier, and filtered online using 60 Hz notch and .1–100 Hz band-pass filters. Scalp electrodes were placed using a 32-channel cap fitted with sintered Ag/AgCl electrodes (Quik-Caps; Neurosoft Inc., El Paso, TX) oriented according to the international 10–20 system, and referenced to the nose-tip. Linked pairs of electrooculogram (EOG) electrodes recorded horizontal (electrodes on the outer canthi) and vertical (electrodes above and below the left eye) eye movements.

9.2. Data processing and artifact filtering

Continuous data was separated into 510 ms epochs. Epochs containing blinks and other artifacts were removed using a maximum voltage criterion of +/−100 μV at either horizontal (HEOG) or vertical electrooculogram (VEOG) electrode channels (pre-artifact =50 ms, refractory period =250 ms, post-artifact =50 ms). Electrode channel signals were also visually inspected, with exceedingly noisy ones eliminated from further analysis. In addition to filtering for artifacts and visually inspecting for noisy channels, we instructed participants to refrain from moving their heads too much (e.g., swaying to the music), but also encouraged them to relax and play naturally. The average number of accepted epochs per condition was 26.95 (SD =3.8) for Pre-stimulus, 276.41 (SD =72.4) for Listen, 300.64 (SD =73.0) for Playback, and 286.86 (SD =83.1) for Improvisation.

Using the t-test for independent samples, we found no significant differences in the number of accepted epochs for Pre-stimulus, Listen, Playback, and Improvisation conditions for the non-FITI (M =25.82, 261.42, 300.67, 278.83, SD =4.1, 89.2, 83.6, 103.1) and FITI (M =28.20, 294.40, 300.60, 296.50, SD =3.3, 42.9, 62.2, 54.1) groups: Pre-stimulus, $t(19)$ = −1.459, $p > .05$; Listen, $t(20)$ = −1.068, $p > .05$; Playback, $t(20)$ =.002, $p > .05$; Improvisation, $t(20)$ = −.487, $p > .05$. Thus, non-FITI and FITI groups did not significantly differ in terms of number of accepted epochs per condition.

Epoch sweeps were next averaged for each condition. Frequency domain averaging was performed wherein amplitude was computed at each frequency band and scaled in terms of power (μV squared, adapted from the Cooley-Tukey method). Frequencies were divided into 1.961 Hz bins from 1 to 250 Hz. Analyses focused on the upper alpha band (10–12 Hz), which is sensitive to group-wise differences in alpha synchronization related to tasks of varying creativity demands (Fink et al., 2009b).

Finally, EEG task related power values were calculated, representing the difference between cortical activity measured at an electrode site during a task interval versus the pre-stimulus reference interval. This subtraction was done for each electrode, for each task (Listen, Playback, Improvisation), for each participant. When reporting task-related

power, decreases from reference are expressed as negative values and referred to as desynchronization, whereas increases from reference are expressed as positive values and referred to as synchronization. Electrodes were grouped as Front Left (FP1, F3, F7) and Front Right (FP2, F4, F8).

10. Assessment of improvised performances

As per Amabile's (1982) Consensual Assessment Technique, we relied on experts in musical improvisation and improvised composition to rate the improvised performances recorded during our EEG testing sessions. The first adjudicator was the vice-president of the Canadian division of a major international record label, the second was a professor of jazz at a college in Toronto, Ontario, and an elite Canadian jazz pianist, and the third was a professor of music at a university in Ontario, and the director of the university's jazz ensemble.

Per Amabile (1982), adjudicators rated independently of one another, and ratings included additional criteria such as technique and aesthetic appeal. These additional criteria provided benchmarks for discriminant validity testing, and broaden the definition of creativity to include aspects of performances and products that signify their overall value. As such, we created the Musical Improvisation Performance Questionnaire (MIPQ) to assess the improvised musical performances recorded during our testing sessions. There were ten questions in this assessment, nine of which were divided into three subscales we called Innovation, Technique, and Musicality, plus an overall impression item. The adjudicators independently rated the improvised performances recorded during the EEG testing sessions at their own home or office.

We asked the experts to score the performances in terms of their universal value (as opposed to in relation to other performances in the sample) and to judge elements of innovation in terms of *goodness* combined with novelty (as opposed to only on novelty). Adjudicators assessed performances using rating scales (1–10) for each question, with one representing a low score and 10 representing a high score. (See appendix A for MIPQ).

11. Results

Participants were categorized into non-FITI and FITI groups based on previous formal institutional training in improvisation. Separate One-way ANOVAs were run to determine differences between non-FITI and FITI groups for each of age, music experience, and improvisation experience. There were no statistically significant differences between groups in terms of age, F (1,20) =.466, p=.503; music experience, F (1,20) =.253, p=.620; or improvisation experience, F (1,20) =.721, p=.406.

12. Task-related differences in alpha synchronization

In the non-FITI group, frontal upper alpha synchronization in the left hemisphere was low during Listen (M =.15, SD =.67, SE =.35), Playback (M =.06, SD =1.00, SE =.43), and Improvisation (M =.25, SD =1.07, SE =.48), whereas frontal upper alpha synchronization in the right hemisphere was high during Listen (M =1.15, SD =2.75, SE =.82), and then moderate during Playback (M =.85, SD =2.46, SE =.83), and during Improvisation (M =.82, SD =2.06, SE =.92). In the FITI group, frontal upper alpha synchronization in the left hemisphere was modest during Listen (M =.48, SD =1.62, SE =.38), increased during Playback (M =.69, SD =1.94, SE =.47), and increased further during Improvisation (M =.87, SD =2.19, SE =.53), while in the right hemisphere it was high during Listen (M =1.19, SD =2.93, SE =.90) increased during Playback (M =1.58, SD =3.28, SE =.90), and then increased to be very high during Improvisation (M =2.51, SD =4.19, SE =1.00).

To test within-group and between-group differences in task related upper alpha (de)synchronization during the music tasks, we employed a two-way repeated measures ANOVA with training group as the between subjects factor and music task and cerebral hemisphere as within subjects factors. The task factor consisted of three levels (Listen, Playback, and Improvisation) and the hemisphere factor of two levels (left and right). The dependent variable was frontal upper alpha synchronization. Mauchly's test indicated that sphericity was assumed, X^2 (2) =3.109, p=.211. The main effects for task, F (2, 40) =1.695, p > .05, and hemisphere, F (1, 20) =3.536, p > .05, were non-significant. Also, the interactions for task by group and hemisphere by group were non-significant F (2, 40) =2.478, p > .05, and F (1, 20) =.084, p > .05, respectively. However, the task-by-hemisphere-by-group interaction was significant, and provided a very large effect size, F (2, 40) =4.425, p < .05, η^2 =.181. Fig. 1 shows task related power changes in frontal upper alpha synchronization in the right hemisphere during the Listen, Playback, and Improvisation tasks for the non-FITI and FITI groups. Note the increase in frontal upper alpha synchronization in the right hemisphere during improvisation which is captured by the three-way interaction noted above.

Post hoc testing using the Fisher LSD test revealed that in the non-FITI group there was no significant difference in frontal upper alpha (de)synchronization in either hemisphere during the Improvisation task in comparison to the Playback task. Also, in the non-FITI group there was no significant difference in frontal upper alpha (de)synchronization in either hemisphere during the Improvisation task in comparison to the Listen task. Conversely, in the FITI group frontal upper alpha synchronization in the right hemisphere was different during the Improvisation task (M =2.51, SD =4.19) compared to during the Playback task (M =1.58, SD =3.28). Furthermore, in this group frontal upper alpha synchronization in the right hemisphere was different during the Improvisation task (M =2.51, SD =4.19) compared to during the Listen task (M =1.19, SD =2.93).

Pearson correlation coefficients were used to examine the relationships between frontal upper alpha synchronization in left and right hemispheres, and between frontal upper alpha synchronization and age and experience within each group. In the non-FITI group there were no statistically significant correlations between frontal upper alpha synchronization in the left and right hemispheres for any tasks (Table 1). Conversely, in the FITI group there were positive correlations between frontal upper alpha synchronization in the left and right hemispheres for all tasks, with all correlations being statistically significant (Table 2). Notably, there were correlations between frontal upper alpha synchronization in the left and right hemispheres during Listen, r(8) =.831, p=.001; Playback r(8) =.847, p=.001; and Improvisation r(8) =.747, p=.006. These findings show that in the FITI group, frontal upper alpha synchronization increased simultaneously across hemispheres during all tasks.

Next, Pearson correlations were run to determine the relationship of

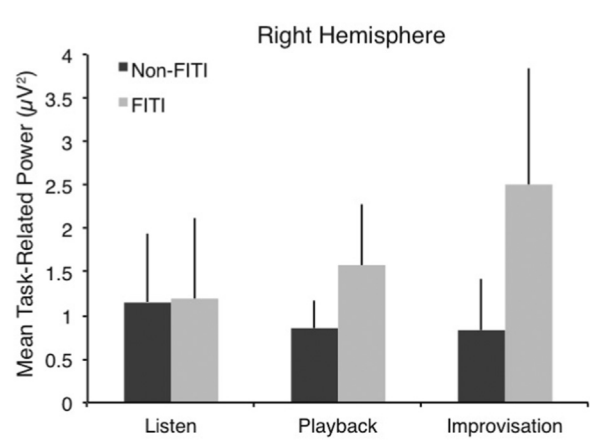

Fig. 1. Frontal upper alpha synchronization during music tasks in the non-FITI and FITI groups.

Table 1
Non-FITI Group (n = 12) Correlations between Frontal Upper Alpha Synchronization in Left and Right Hemispheres, by Experimental Task.

	Right Listen	Right Play	Right Improvise
Left Listen	.040	.104	−.026
Left Play	−.177	.214	.175
Left Improvise	−.413	.079	.104

Note. Correlation coefficients are provided above. For each hemisphere, tasks are presented from left to right in order of increasing creative demands. FITI: formal institutional training in improvisation.

Table 2
FITI Group (n = 10) Correlations between Frontal Upper Alpha Synchronization in Left and Right Hemispheres, by Experimental Task.

	Right Listen	Right Play	Right Improvise
Left Listen	**.831****	**.832****	**.755****
Left Play	**.825****	**.847****	**.739****
Left Improvise	**.761****	**.780****	**.747****

Note. Correlation coefficients are provided above. For each hemisphere, tasks are presented from left to right in order of increasing creative demands. FITI: formal institutional training in improvisation,
** = p < .01.

frontal upper alpha synchronization in the left and right hemispheres with participant age, music experience, and improvisation experience. In the non-FITI group, there were no significant correlations between frontal upper alpha synchronization in the left hemisphere during Listen, Playback, and Improvisation and any of participant age, music experience, and improvisation experience. However, there were a number of statistically significant negative correlations in the right hemisphere (Table 3). These correlations were largest between Improvisation and music experience, $r(10) = -.559$, $p = .029$, and between Improvisation and improvisation experience, $r(10) = -.579$, $p = .024$. As shown in Table 3, there were also negative correlations between frontal upper alpha synchronization in the right hemisphere during Listen, Playback, and Improvisation, and age, music experience, and improvisation experience that were significant.

In the FITI group, there were statistically significant positive correlations between frontal upper alpha synchronization in the left hemisphere during all tasks, and age, music experience, and improvisation experience (see Table 4). However, there were no statistically significant correlations for right hemisphere upper alpha synchronization and any of tasks, age, music experience, and improvisation experience. Correlations of musical and improvisation experience with frontal right upper alpha power for both FITI and non-FITI participants are presented in Figs. 2 and 3. Note the significant inverse correlations in the non-FITI group.

13. Assessment of improvised performances

13.1. Assessment reliability

The intra-class correlation coefficient (ICC) measured inter-rater reliability (IRR) for expert adjudicators of improvised performances using the Musical Improvisation Performance Questionnaire (MIPQ). Here, inter-rater reliability is akin to construct validity following Amabile's (1982) notion that a product is deemed creative as much as it is agreed to be so by appropriate observers. As such, Intra-class Correlations were computed for absolute agreement. First, IRR was calculated for Innovation ($ICC = .76$, $N = 3$), Technique ($ICC = .81$, $N = 3$), and Musicality ($ICC = .77$, $N = 3$). Next, IRR was calculated for Subscales Total ($ICC = .80$, $N = 3$), and Full Total ($ICC = .81$, $N = 3$). All scales and totals had reliability in the 'excellent' range (i.e., $ICC > .75$; Fleiss, 1986).

Given high inter-rater agreement in scoring of subscales, subscale totals, and overall measure totals, we averaged the scores for the three adjudicators to yield composite subscale, subscale total, and measure total scores for subsequent analyses.

13.2. Between-group comparisons in performance

Subscales of the MIPQ are presented in Table 5. Scores were very similar across the non-FITI and FITI groups, confirmed by a failure to note significant differences between groups on Innovation, $t(20) = -.692$, p= > .05; Technique, $t(20) = -.620$, p > .05, Musicality, $t(20) = -.696$, p > .05, subscales total, $t(20) = -.696$, p > .05, or overall total, $t(20) = -.696$, p > .05.

13.3. Correlations of MIPQ subscales with task-related alpha synchronization

Pearson correlations were used to assess the relationships between MIPQ subscales, subscale totals, and full measure totals and frontal upper alpha synchronization in the right hemisphere during Listen, Playback, and Improvisation for the non-FITI and FITI groups (Tables 6, 7; Fig. 4). In the non-FITI group there were no significant correlations between any of the music rating measures and frontal upper alpha synchronization in the right hemisphere during any of Listen, Playback, or Improvisation. Conversely, the FITI group showed positive correlations between the MIPQ measures and frontal upper alpha synchronization in the right hemisphere during Improvisation. Correlations between right frontal upper alpha synchronization during Improvisation and both the Innovation, and Technique subscales just missed significance ($r(8) = .543$, $p = .052$, and $r(8) = .539$, $p = .054$, respectively).

14. Discussion

The primary aim of this study was to investigate differences in frontal brain activity occurring between tasks with high and low

Table 3
Non-FITI Group (n = 12) Correlations of Frontal High Alpha Synchronization During Music Tasks with Age, Music Experience, and Improvisation Experience, by Cortical Hemisphere.

	Left Hemisphere			Right Hemisphere		
	Listen	Playback	Improvisation	Listen	Playback	Improvisation
Age	−.025	.113	.006	−.421	−.290	−.356
MUS exp	−.087	−.082	.010	−.479	**−.528***	**−.559***
IMPRV exp	.006	.014	.099	**−.514***	−.471	**−.579***

Note. Correlation coefficients are provided above. For each hemisphere, tasks are presented from left to right in order of increasing creativity demands. FITI: formal institutional training in improvisation * = p < .05.

Table 4
FITI Group (n = 10) correlations of frontal high alpha synchronization during music tasks with age, music experience, and improvisation experience, by cortical hemisphere.

	Left Hemisphere			Right Hemisphere		
	Listen	Playback	Improvisation	Listen	Playback	Improvisation
Age	**.758****	**.739****	**.711***	.471	.450	.390
MUS exp	**.745****	**.738****	**.752****	.538	.515	.515
IMPRV exp	**.795****	**.761****	**.784****	.549	.537	.542

Note. Correlation coefficients are provided above. For each hemisphere, tasks are presented from left to right in order of increasing creativity demands. FITI: formal institutional training in improvisation
* = p < .01.

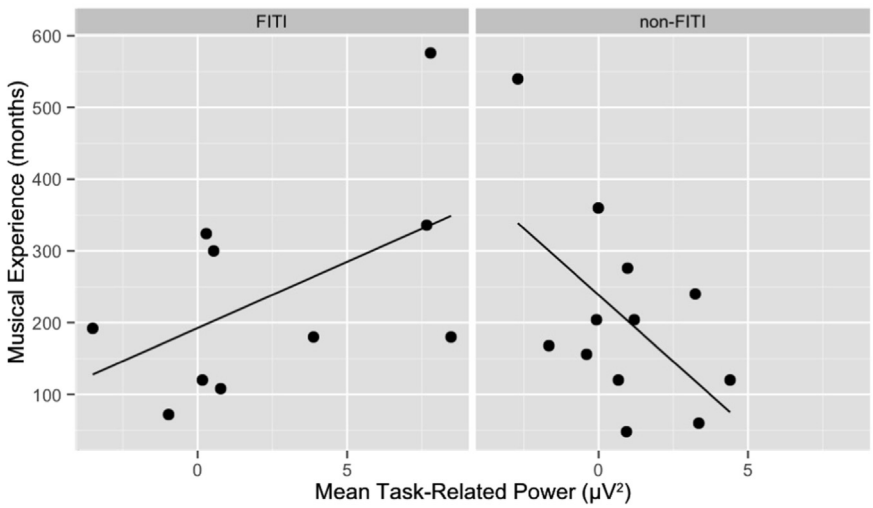

Fig. 2. Correlations of mean musical experience (months) with right frontal upper alpha task-related power (μV^2) for FITI and non-FITI groups.

creativity demands. We examined this by recording EEGs while musicians listened to, played back, and improvised jazz melodies. It was hypothesized that musicians would show increased frontal upper alpha band activity during more creative tasks compared to during less creative tasks, and that this effect would be more evident in those with previous Formal Institutional Training in Improvisation (FITI) than those without. We also aimed to investigate whether frontal upper alpha activity is associated with higher quality improvised performances, and to see if this effect would be more evident in those with previous FITI than those without.

We first considered the theory that creativity is a distinct, mental state that benefits from formal training. Our results showed that frontal upper alpha synchronization increased as a function of an interaction between task, hemisphere, and FITI groups. Follow-up analyses revealed significant task-related differences between frontal upper alpha synchronization during Improvisation and Playback, and during Improvisation and Listen. This showed that amongst those with previous FITI, frontal upper alpha synchronization was greater

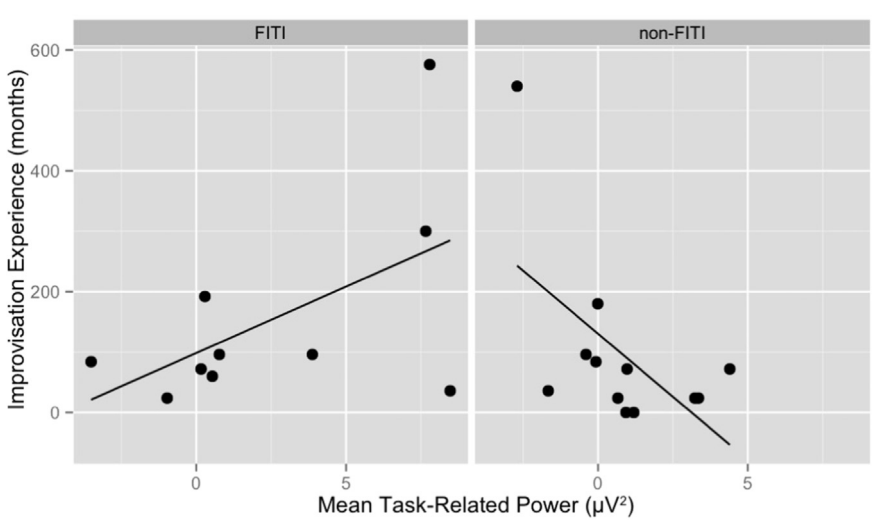

Fig. 3. Correlations of mean improvisation experience (months) with right frontal upper alpha task-related power (μV^2) for FITI and non-FITI groups.

Table 5
Means and Standard deviations for Subscales, Subscale Total, and Full Total of the Musical Improvisation Performance Questionnaire.

	Innovation	Technique	Musicality	Sub Total	Full Total
Non-FITI	4.6 (1.33)	4.3 (1.58)	4.4 (1.55)	4.4 (1.47)	4.3 (1.46)
FITI	4.9 (1.28)	4.7 (1.53)	4.8 (1.56)	4.8 (1.45)	4.8 (1.50)
Total	4.7 (1.29)	4.5 (1.53)	4.6 (1.54)	4.6 (1.45)	4.5 (1.46)

Table 6
Non-FITI Group (n = 12) Correlations between the Musical Improvisation Performance Questionnaire Subscales, Subscale Total, and Full Total and Frontal High Alpha Synchronization in the Right Hemisphere during Music Experiment Tasks.

	Listen	Playback	Improvise
Innovation	.147	.030	−.007
Technique	.179	.063	−.015
Musicality	.108	.016	−.074
Subscale Total	.146	.037	−.033
Full Total	.117	.018	−.053

Note. No significant correlations at the .05 level in this table.

Table 7
FITI Group (n = 10) Correlations between the Musical Improvisation Performance Questionnaire Subscales, Subscale Total, and Full Total and Frontal High Alpha Synchronization in the Right Hemisphere during Music Experiment Tasks.

	Listen	Playback	Improvise
Innovation	.462	.423	.543
Technique	.446	.403	.539
Musicality	.488	.448	**.592***
Subscale Total	.467	.426	**.560***
Full Total	.461	.422	**.561***

Note. Bold type with * denotes significance at the .05 level.

while musicians improvised than while they played back melodies and even greater than while they just listened to melodies. This finding supports dual-process models involving intuitive Type 1 creative thinking, Dietrich's (2004a) theory that alpha activity signifies spontaneous processing, and Fink et al.'s (2009b) findings that showed alpha synchronization occurring during the spontaneous composition of dance pieces. Thus, we interpret the increased frontal upper alpha synchronization that occurred while musicians improvised melodies as evidence of an underlying creative mental state characterized by immersion in a Type 1 spontaneous processing mode.

Increased frontal alpha during creative ideation was at first assumed to signify reduced cortical activity characterized by the transient suppression of executive functions and logical-rational thought processes (Dietrich, 2003; Fink and Neubauer, 2006), sometimes described as *cortical idling* (Pfurtscheller, 1999; Pfurtscheller et al., 1996). However, more recently researchers have suggested increased frontal alpha is instead indicative of top-down active processes including sensory inhibition and internal focus of attention (Buzsáki and Draguhn, 2004; Klimesch et al., 2007; Ward, 2003). Creativity researchers have continued testing the role of alpha during ideation and have found it to reflect active internal processing mechanisms as well (Benedek et al., 2011; Fink et al., 2009a). For example, Fink et al. (2009a) used both EEG and fMRI to investigate the role of internal processing mechanisms as they relate specifically to creative ideation and interpreted their findings to strongly suggest "that alpha synchronization during creative ideation reflects an active cognitive process rather than cortical idling" (Fink and Benedek, 2014, p. 7). Considering these findings, the task-related differences in upper alpha synchronization between tasks of high and low creativity demands observed in our study, likely as well suggest top-down processing and internal focus of attention, and not cortical idling.

Advancing understandings of spontaneous processing further, synthesis of current fMRI research has led to an emerging theoretical framework explaining frontal deactivation as implicated in a default-control network coupling involving concurrent generative and evaluative processing (Beaty et al., 2016). This may suggest a more complex interplay of processes involving both bottom-up generative and at least some degree of top-down evaluative monitoring. Since collected fMRI research shows that the degree to which tasks are goal-directed modulates frontal (de)activation (Beaty et al., 2016), future EEG research delineating several generative artistic performance tasks with varying degrees of high to low goal-direction is recommended.

The results of the present study showed that amongst those with previous FITI there was significantly and substantially greater frontal upper alpha synchronization in the right hemisphere during improvisation, compared to during deliberate playback. This effect was greatest in right hemisphere electrodes. These results speak to the prior literature, which has not conclusively established that alpha activity is hemisphere-selective during creative cognition (Dietrich and Kanso, 2010). At best, the literature so far might indicate a special

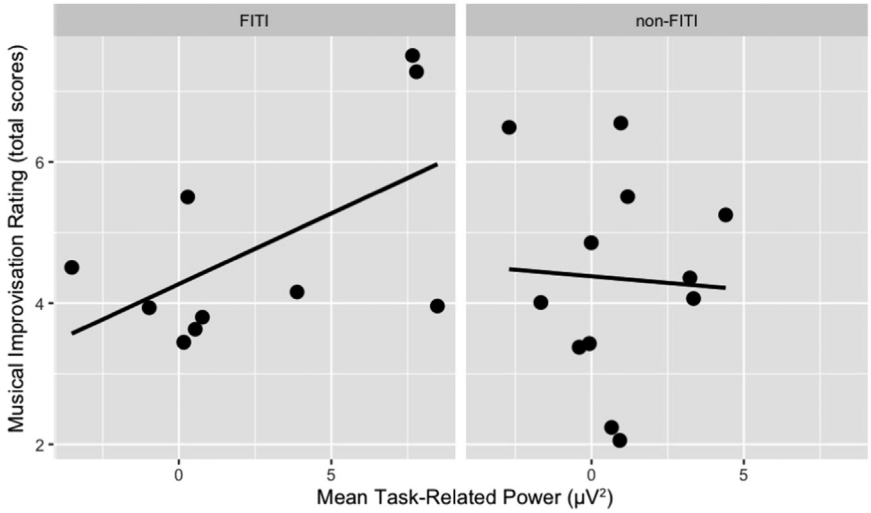

Fig. 4. Correlations of mean musical improvisation performance ratings (0–10 pts) with right frontal upper alpha task-related power (μV^2) for FITI and non-FITI groups.

role for the right hemisphere as related to creative ideation (Bhattacharya and Petsche, 2002, 2005; Fink et al., 2011, 2009b; Grabner et al., 2007; Martindale et al., 1984; Petsche et al., 1997). Overall, despite contemporary concerns about attributing creative cognition processes to one hemisphere or the other, the increased right hemisphere effects we found during musical improvisation support the view of a special role of right frontal brain areas in the generation of original ideas (Fink and Neubauer, 2006; Grabner et al., 2007) and as benefitting from expertise and development through training (Fink et al., 2011; Rosen et al., 2016). We do note, however, that upper alpha synchronization levels increased for both hemispheres during the creativity manipulation; specifically, the effect was larger in the right than left hemisphere for the previous-FITI group. One explanation is that the smaller increased left hemisphere synchronization reflects interhemispheric interactions during creativity, with the left hemisphere playing a secondary role in this regard (Lindell, 2011).

Amongst participants without previous FITI, right frontal upper alpha synchronization was moderately negatively correlated with both music and improvisation experience. This effect was strongest during the improvisation sub-task. Conversely, those with previous FITI showed no correlations here. We interpret this difference as reflecting the non-previous-FITI group's lack of immersion in Type 1 spontaneous processing such that as these musicians gain experience they become set in processing musical ideas deliberately rather than spontaneously. The non-FITI musicians' lexicons of musical phrases and theoretical knowledge may increase over time, however they may approach performance from a more deliberate and less spontaneous processing mode as shown in Rosen et al.'s (2016) tDCS research. This finding may further suggest that in general, spontaneous processing is an ability or skill that can be developed through intentional instruction in creative process, here operationalized as musical improvisation within a harmonic framework.

We ran correlations between right frontal upper alpha synchronization during improvisation and expert assessments of performance quality to determine the relationship between musicians' neural activity during spontaneous composition and the quality of the products created thereupon. We hypothesized that frontal upper alpha synchronization would positively correlate with MIPQ scores. Accordingly, the non-FITI group showed no significant correlations between any of the MIPQ measures and right frontal upper alpha synchronization occurring while musicians listened to, played back, or improvised melodies. Conversely, in the FITI group there were several positive medium-sized correlations between right frontal upper alpha synchronization and MIPQ measures including correlations with Musicality, subscales total scores, and overall full measure scores. These correlations indicate that in our sample right frontal upper alpha synchronization was associated with the quality of improvised performances for those with previous FITI. These findings are in accord with previous research that has shown frontal alpha synchronization to be related to the originality of ideas generated during divergent thinking tasks (Fink and Neubauer, 2006; Grabner et al., 2007).

In sum, the correlations of right frontal upper alpha synchronization and MIPQ scores suggest that for those with previous FITI, immersion in Type 1 spontaneous processing tends to yield higher quality improvised performances that are rated as being better overall by domain experts.

15. Implications, limitations, and future research

The sample for this study consisted of 22 musicians who were categorized into non-FITI and FITI groups of 12 and 10 participants. We found differences between groups, however there may have been more subtle effects in the data that were simply not detectable with modest sample sizes. Thus, readers should exercise a degree of caution when reading into null effects such as the failure to find an effect in one of the groups. Rather, we emphasize that there is a relationship, and that the effect is bigger in one group than the other. Additionally, this study solely examined creativity in musicians. It remains to be seen how they will generalize to other types of artists (e.g., writers, visual artists, dancers) and other types of creative people (e.g., mathematicians, scientists, engineers, computer programmers), given key differences in the creative works that are generated by these different groups of individuals.

The primary data collection method that we employed was EEG. Although EEG has high temporal accuracy it is limited in its spatial accuracy and in its ability to isolate specific areas over the scalp given that it is a measure of electrical current at the surface of the scalp after conductance through cranial matter that varies in density. This introduces spatial distortions that limit our ability to infer the source of these signals (Arden et al., 2010). Thus, the neural localization of our effects may be limited. Further work is recommended to replicate our research using spatially precise complementary approaches such as MEG or fMRI.

16. Conclusion

In the present study, we contribute to creative cognition literature by measuring frontal activity while contrasting tasks biased towards deliberate and spontaneous creative processing. We hypothesized that musicians would show increased frontal upper alpha band activity during tasks with high creative demands compared to those with low creative demands, and that this effect would be more evident in individuals with previous FITI than those without. Confirming this hypothesis, individuals with previous FITI showed greater upper alpha synchronization during spontaneous composition versus deliberate playback and versus listening to melodies passively. We interpreted this as evidence of a creative mental state characterized by spontaneous processing, and likely by a degree of top-down inhibition and internal focus of attention–characteristics implicated in both competitive and concurrent interactions of default and control networks during generative artistic performance. Further, we hypothesized that immersion in such a creative mental state would be associated with higher quality creative products, and that this effect would be more evident in those with previous FITI than those without. Confirming this hypothesis, for individuals with previous FITI, upper alpha synchronization was associated with higher quality performances as rated by domain experts.

Overall, the task-related differences in upper alpha synchronization that we observed support the view that creativity can be conceptualized as a distinct mental state occurring during spontaneous musical performance. Our findings further imply that creativity can be nurtured through training, and illustrate that immersion in the creative state has high cultural and economic value because it yields higher quality products.

Acknowledgment

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Appendix A. – Musical Improvisation Performance Questionnaire (MIPQ)

Music Adjudication Questionnaire Participant #_____

Innovation

1. How unique is this improvisation compared to the melody?

   1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10

2. How unique is this improvisation compared to what you'd expect someone to play over these chords?

   1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10

3. How unique/novel is this improvisation overall?

   1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10

Technical Ability

4. How much command of the instrument is displayed?

   1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10

5. How complex are the musical thoughts/ideas?

   1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10

6. How theoretically advanced is this improvisation?

   1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10

Musicality

7. How enjoyable or fulfilling are the tonal and rhythmic sequences played?

   1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10

8. How much personal style is present in the improvisation?

   1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10

9. How clear are the musical ideas in this improvisation?

   1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10

Overall impression

10. How would you rate the quality of this improvisation overall?

    1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10 1 2 3 4 5 6 7 8 9 10

Appendix B. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.neuropsychologia.2017.03.020.

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